

MUHAMMAD FAREED

Date of Birth: 15 August 1994 | Nationality: Pakistani | Address: Lahore, Pakistan
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RESEARCH INTERESTS

Efficient, privacy-preserving medical AI for resource-constrained clinical environments; Vision–Language Models and multimodal fusion for automated clinical decision support; parameter-efficient fine-tuning (QLoRA/LoRA) and quantization for on-premise healthcare deployment; systematic evidence synthesis in healthcare AI; ethical governance of clinical AI systems.

EDUCATION

MS in Data Science | 2024 – 2026 (CGPA: 4.0/4.0)

Riphah School of Computing & Innovation (RSCI), Riphah International University, Lahore Campus

- First position (Spring 2024 cohort) | Granted *Dean's Honor Award (Certificate of Merit Achievement)*
- Thesis: "Retrieval-Augmented, Parameter-Efficient Fine-Tuning of a Vision–Language Model for Chest X-ray Report Generation" | Supervisor: Dr. Jamal Uddin
- Relevant coursework (A+ = **, A = *): Machine Learning**, Research Methods**, Advanced Database Systems**, Cloud Computing Infrastructure & Services**, Professional Ethics-I**, Deep Learning*, Statistical & Mathematical Methods for Data Science*, Tools & Techniques in Data Science*, Big Data Analytics*

BS (Hons) in Mathematics | 2012–2016 (First Division)

University of Education, Township Campus, Lahore

B.Ed. Secondary (1.5-Year), Specialisation: ICT in Education | 2021–2024 (First Division)

Virtual University of Pakistan, Lahore Campus

F.Sc Pre-Engineering (HSSC) | 2010–2012 (Grade A – 100% merit scholarship)

Punjab College of Science, Muslim Town Campus, Lahore

Secondary School Certificate — Science (SSC) | 2008–2010 (Grade A+ – First position at school level)

Aziz Memorial Science High School, Lahore

PUBLICATIONS

- Fareed, M.**, Fatima, M., Uddin, J., Ahmed, A., & Sattar, M.A. (2025). "A Systematic Review of Ethical Considerations of Large Language Models in Healthcare and Medicine." *Frontiers in Digital Health*. IF 3.8, Q1. <https://doi.org/10.3389/fdgth.2025.1653631>
- Fareed, M.**, Uddin, J., Ahmed, A., Sayyid, A.H., & Sattar, M.A. (2026). "CXRG-SVLM: A Parameter-Efficient Small Vision-Language Model for Multi-View Chest X-ray Report Generation in Resource-Constrained Environments." *Scientific Reports* (under review, MS thesis work)
- Fareed, M.**, Uddin, J., Ahmed, A., Nauman, M.A., Khan, N., Billah, M., & Sattar, M.A. (2026). "A Systematic Review of LSTM-Based Deep Learning Models in Healthcare and Medicine." *Plos One* (under review)
- Ahmed, A., Riaz, S., & **Fareed, M.** (2026). "Optimizing Multilingual BERT for Punjabi Sentiment Analysis using PEFT Techniques: Balancing Accuracy and Parameter Efficiency." *Computers, Materials & Continua (CMC)*, Tech Science Press, IF 1.7 (2024), Scopus CiteScore 6.6 (2025) (under review)

RESEARCH & TECHNICAL PROJECTS

1. Roman Urdu Small Language Model (RU-SLM)

- Applied 4-bit QLoRA to an 8B-parameter model for conversational Roman Urdu NLP under severe compute and data constraints.
- Achieved ~74% reduction in trainable parameters with retained deployable inference quality.

2. Text Summarisation Small Language Model (TS-SLM)

- Fine-tuned Mistral-7B with LoRA and 4-bit quantisation for length-controlled abstractive news summarisation.
- Enforced explicit output-length constraints at inference; measured faithfulness vs. hallucination via ROUGE and BERTScore.
- Mapped compression–quality trade-offs across PEFT configurations on generation tasks.

3. Multilingual Sentiment and Language Detection

- Fine-tuned Gemma-3 and Qwen-3 for joint language identification and sentiment classification across Urdu and Roman Urdu.

- Structured outputs as validated JSON for downstream pipeline integration.
- Documented distinct error modes between script-identical and transliterated varieties; informs low-resource multilingual model design.

Additional projects at: github.com/mfareedkhan

AWARDS & HONOURS

1. Selected for the inaugural cohort, AI-Driven Enterprise Product Development Programme — Prime Minister's Youth Programme (PMYP) × Stanford Engineering CGOE, Oct 2025. (Competitive national selection; cohort developed and deployed AI-driven enterprise solutions under Stanford faculty guidance.)
2. Best Teaching Performance Award — PEN NGO, Lahore, Dec 2024.
3. 100% merit scholarship, F.Sc Pre-Engineering — Punjab College of Science, Lahore, 2010–2012.
4. First position at school level, Secondary School Certificate (Science, Grade A+) — Aug 2010.
5. Prime Minister's Laptop Scheme — awarded on academic merit, 2012.

SKILLS & TECHNICAL EXPERTISE

- **Programming:** Python (primary), LaTeX
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers & PEFT, Keras
- **Efficient Adaptation & Deployment Methods:** QLoRA, LoRA, 4-bit Quantization
- **Modelling & Research Methods:** Vision–Language Models (VLMs, Large Language Models (LLMs), Multimodal Fusion, Transfer Learning, Parameter-Efficient Fine-Tuning, NLP, Computer Vision, Generative AI, Systematic Literature Review (PRISMA 2020 · Kitchenham & Charters 2007)
- **Infrastructure & Data:** Cloud Computing, SQL & NoSQL Databases, Google Colab, Kaggle Notebooks

TEACHING & PROFESSIONAL EXPERIENCE

- 2018–present **Science & Mathematics Teacher (BPS-14*)**, Govt. Elementary School, Attari Saroba, Lahore — *held concurrently with MS studies, thesis research, and four peer-reviewed papers (2024–2026)*
- 2022–2023 **Digital Marketing & Web Development Trainer**, Digiversity Training Institute, Lahore
- 2021–2022 **Social Media Marketing Manager**, Conversion Marketing Agency, Lahore
- 2015–2017 **Mathematics & Physics Instructor** (intermediate & undergraduate), Leads Group of Colleges, Lahore
- 2014–2015 Private **O/A-Level Mathematics Tutor**
- 2012–2014 **Science & Mathematics Teacher** (matriculation), Yasin Public High School, Lahore
- * **BPS-14:** substantive permanent post in Pakistan's national civil service grading structure, awarded by competitive examination.

VOLUNTEER WORK

- 2018–2020 **Vice President**, Tameer Kadah Foundation (TKF), Lahore — welfare and development initiatives for children with physical and cognitive disabilities; co-managed organisational programmes and community outreach.

CERTIFICATIONS & TRAINING

- **Innovation-Driven Product Development Using AI** | Feb 2026 | *Stanford Engineering CGOE*
- **Deep Learning Specialization (5 courses)** | Sep–Dec 2025 | *DeepLearning.AI*
- **Generative AI Engineering and Fine-tuning Transformers** | Jul 2025 | *IBM*
- **Data Science Ethics** | Sep 2025 | *University of Michigan*

SEMINARS, CONFERENCES & WORKSHOPS

- COLLOQUIUM: Beyond the Cloud — Shaping the Next Era of Computing | Jan 2026 | *Centre for Research in Intelligent, Distributed & Secure Computing × RSCI, Lahore*
- ICCI 2025 — International Conference on Computing & Innovation, RSCI, Lahore | Dec 2025 | Competed in 3MT (Three-Minute Thesis) Competition
- Poster Presentation, RSCI Research Week | Sep 2025 | *RSCI, Lahore*
- Online Seminar on Research Trends and Advances, RSCI | May 2024

PROFESSIONAL TEACHING TRAININGS

- *Six-Week Higher-Order Thinking Skills Programme* | Dec 2025 | *PECTAA, Lahore*
- *Four-Week Induction Training Programme* | Feb 2018 | *QAED, Lahore*

LANGUAGE PROFICIENCY

- English — Professional working proficiency (C1) [formal test pending — update with IELTS/TOEFL score]
- Urdu — Native (C2) | Punjabi — Native (C2)

REFERENCES

Dr. Jamal Uddin

Assistant Professor; Head of Dept., AI & Data Science

Riphah School of Computing & Innovation (RSCI), Riphah International University, Lahore Campus

Email: jamal.din@riphah.edu.pk

Relationship: MS thesis supervisor; taught Deep Learning and Statistical & Mathematical Methods for Data Science

Dr. Muhammad Yaseen

Associate Professor; Head of Dept., Software Engineering; In-Charge Graduate/Postgraduate Programs

Riphah School of Computing & Innovation (RSCI), Riphah International University, Lahore Campus

Email: muhammad.yaseen@riphah.edu.pk

Relationship: Taught Advanced Database Systems and Cloud Computing Infrastructure & Services

Dr. Muhammad Awais Sattar

Postdoctoral Researcher, Dept. of Computer Science, Electrical & Space Engineering

Luleå University of Technology, Sweden

Email: muhammad.awais.sattar@associated.ltu.se

Relationship: Taught Tools & Techniques in Data Science, Programming Fundamentals, and Research Methods; guided graduate research; corresponding author on Papers 1 and 3